AI Use and Employment Change Across Industries

1. Background

Artificial intelligence (AI) may help businesses complete tasks faster, change the skills workers need, or affect employment. Its effects may differ by industry. Previous research also suggests that AI can support or change jobs rather than simply replace them (Georgieff & Hyee, 2021; OECD, 2023; Gmyrek et al., 2023). This makes it useful to compare AI use and employment across industries and countries.

2. Research Aim and Questions

This study examines how business AI use and employment changed across industries. It asks: (1) How did AI use change from 2023 to 2025? (2) How did employment change over the same period? (3) Do industries with larger increases in AI use also show larger employment changes? (4) Are these patterns similar across countries?

For the midterm, Denmark is the representative country case. The final project will add countries with comparable data to see whether the patterns are similar.

3. Data Sources and Measures

Both datasets were downloaded from the OECD Data Explorer: ICT Access and Usage by Businesses and Annual Employment by Detailed Economic Activity, Domestic Concept ([4], [5]). The data were matched by industry and year, giving 27 observations across nine industries and three years. AI use is the share of businesses using AI; employment is the number of people employed. AI-use change is measured in percentage points, while employment change is measured as a percent change.

The nine industries include manufacturing, construction, wholesale and retail trade, transport

and storage, accommodation and food services, information and communication, professional and scientific activities, and administrative and support services.

4. Methods, Visualizations, and Preliminary Findings

Figures 1 and 2 compare AI use and employment across industries. AI use increased in all nine industries. Professional and scientific activities had the largest increase (about 40.13 percentage points), while information and communication had the highest 2025 AI-use rate (about 79.08%). Employment increased in eight industries from 2023 to 2025 and declined in information and communication. The largest increases were in professional and scientific activities, administrative and support services, and manufacturing.

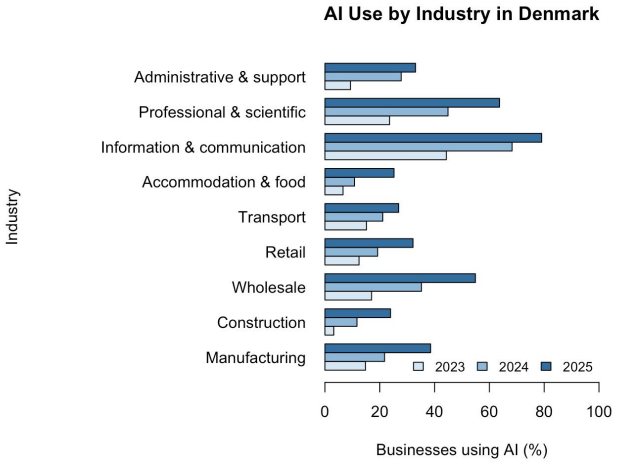


Figure 1. AI use by industry and year.

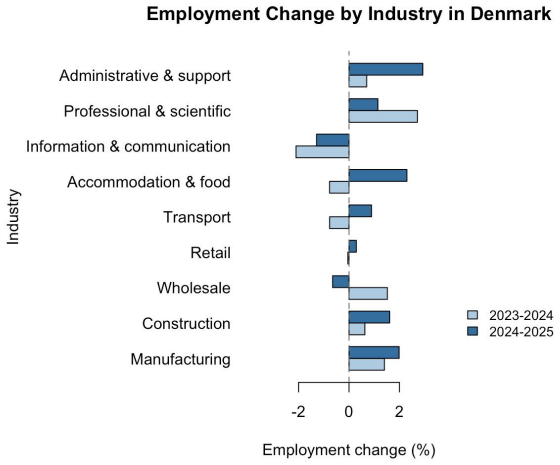


Figure 2. Employment change by industry.

Figure 3 compares the change in AI use with employment change from 2023 to 2025. The pattern differs by industry: employment rose in professional and scientific activities but fell in information and communication, even though AI use increased in both.

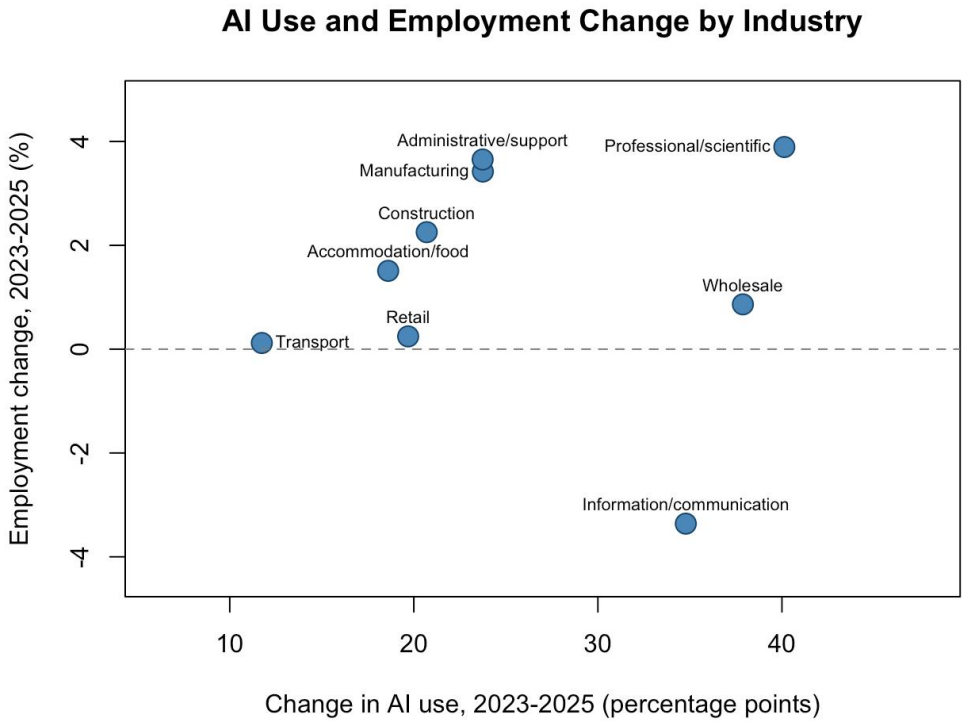


Figure 3. AI-use change and employment change by industry, 2023–2025.

5. Limitations

This descriptive analysis cannot establish that AI caused employment changes; economic conditions, demand, company growth, and other technologies may also matter. The midterm covers only Denmark, nine industries, and three years. Employment totals do not show individual workers' job changes, and the AI measure does not show how often or effectively businesses use AI.

6. Conclusion and Next Steps

The Denmark results differ across industries. The final project will add countries with comparable data and use the same measures where possible to check whether these patterns also appear elsewhere.

7. References

[1] Georgieff, A., & Hyee, R. (2021). Artificial intelligence and employment: New cross-country evidence. OECD Social, Employment and Migration Working Papers, No. 265. OECD publication

[2] OECD. (2023). OECD Employment Outlook 2023: Artificial Intelligence and the Labour Market. OECD Publishing. OECD publication

[3] Gmyrek, P., Berg, J., & Bescond, D. (2023). Generative AI and Jobs: A global analysis of potential effects on job quantity and quality. International Labour Organization. ILO publication

[4] OECD. ICT Access and Usage by Businesses. OECD Data Explorer. Dataset

[5] OECD. Annual Employment by Detailed Economic Activity, Domestic Concept. OECD Data Explorer. Dataset